

TAN XIU LI, FELICIA

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PROFILE

Year 2 Diploma in Information Technology student at Nanyang Polytechnic and Director's List recipient with hands-on experience in software engineering, Edge AI, secure systems and systems integration. Member of the EchoSync team selected as a Top 5 Finalist out of 81 teams from 17 institutions in the SCDF & Dell Lifesavers' Innovation Challenge. Experienced in real-time sensor processing, biometric access control and DevSecOps, with an interest in applying these capabilities to Singapore's defence and national security needs.

EDUCATION

Diploma in Information Technology

Nanyang Polytechnic | Apr 2025 – Expected Apr 2028

Cumulative GPA: 3.74/4.00

Higher National ITE Certificate in IT Application Development

Institute of Technical Education | Jan 2022 – Dec 2024

Cumulative GPA: 3.639/4.00

TECHNICAL SKILLS

- Programming & Frameworks: Python, JavaScript, React, Next.js, Node.js, Express.js, Flask, FastAPI
- Data & Systems: SQL, SQLite, MongoDB, REST APIs, Real-Time Data Processing, Systems Integration
- AI & Security: Edge AI, Computer Vision, Facial Recognition, RBAC, Audit Logging
- DevSecOps & Testing: GitLab CI/CD, Pytest, Selenium, Postman, SAST, Dependency scanning and Secret Detection

KEY TECHNICAL PROJECTS

EchoSync | React, Next.js, Python, Arduino, Raspberry Pi, Edge AI Mar 2026 – Jul 2026

- Engineered an AI-assisted emergency response prototype selected as a Top 5 Finalist in the SCDF & Dell Lifesavers' Innovation Challenge.
- Integrated real-time sound, motion, distance and load sensor data through Arduino and Raspberry Pi edge devices to detect potential falls, distress and unusual inactivity.
- Developed privacy-conscious sensor-processing and emergency alert-routing workflows connecting an operational dashboard, caregiver application and community responder escalation interface.
- Designed explainable risk assessments and multilingual voice check-ins to support timely emergency-response decisions.

FlowGuard | React, Node.js, Express.js, MongoDB, FastAPI

Apr 2026 – Present

- Developed a full-stack AI facility monitoring and access-management system supporting secure operations across multiple facility zones.
- Implemented role-based access control and facial-recognition enrolment for identity verification, gate access and attendance tracking.
- Integrated the React frontend, Node.js backend, FastAPI recognition service and edge-camera components through REST APIs.
- Developed audit logging, access-event tracking and real-time dashboards to improve accountability and operational awareness.

SwapLah | Python, Flask, SQLite, GitLab CI/CD

Apr 2026 - Present

- Developed a secure campus marketplace featuring authentication, session management, role-protected administrative functions, listings, offers and reviews.
- Implemented server-side validation, password hashing, parameterised database operations and protected API routes.
- Built automated unit, API and Selenium UI tests with a 75% coverage gate in a multi-stage GitLab CI/CD pipeline.
- Applied DevSecOps controls including SAST, dependency scanning, secret detection and automated security gates.

AWARDS & CCA ACHIEVEMENTS

- Top 5 Finalist, SCDF & Dell Lifesavers' Innovation Challenge — Project EchoSync, 2026
- Director's List, Nanyang Polytechnic — Sem 2 AY2025/2026
- Director's List, Institute of Technical Education — 2022 and 2024
- Edusave Skills Award, 2024
- Edusave Merit Bursary, 2022–2024
- MOE EAGLES Award, 2022
- ITE Sports Council Badminton Championship — Gold, 2022 and 2023; Silver, 2024

CERTIFICATIONS & TECHNICAL TRAINING

- Generative AI & Machine Learning — AWS
- Alibaba Cloud Fundamentals — Alibaba Cloud
- User Experience Design Fundamentals — IBM
- Web Development Fundamentals — IBM
- Sandbox Integration for QR and eNETS — NETS

WORK EXPERIENCE

W360 Group Pte Ltd

Digital Marketing Executive, Contract | Apr 2024 – Jul 2024

IT & Digital Content Intern | Sep 2023 – Apr 2024

- Maintained client websites and implemented interface, content and layout updates based on project requirements.
- Produced website mockups and UI prototypes and translated client feedback into scoped development tasks.
- Conducted interface checks, content deployment, SEO updates and post-launch revisions.